

EXPERIMENT-14

14. | Write a code using RAW sockets to implement packet sniffing.

packet_sniffer.py

```
import socket

HOST = socket.gethostbyname(socket.gethostname())

sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)

sniffer.bind((HOST, 0)) sniffer.setsockopt(socket.IPPROTO_IP,
socket.IP_HDRINCL, 1)

print("Packet Sniffer Started")

print("Listening on:", HOST) print("Press
Ctrl+C to stop")

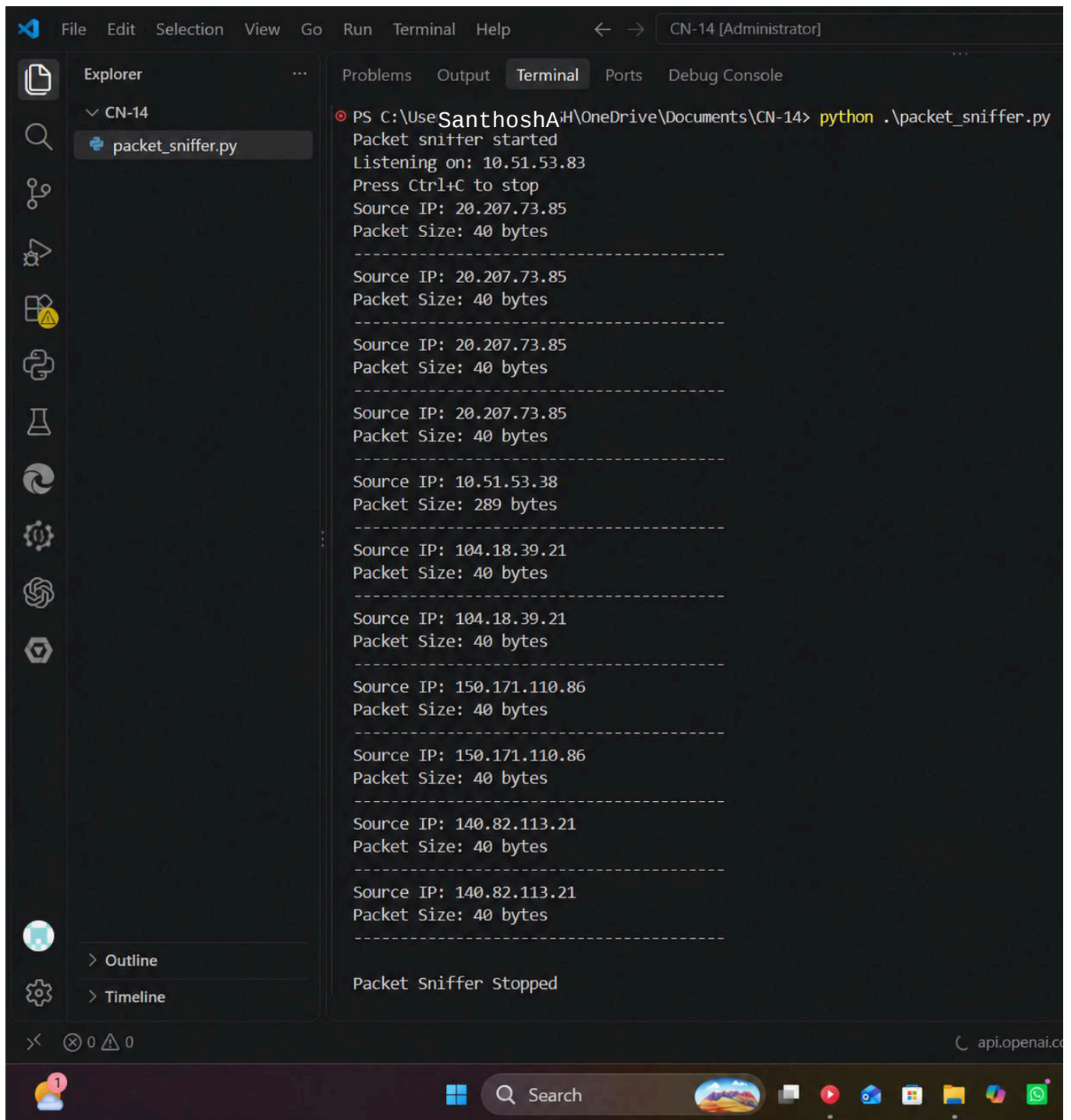
try:
    while True:
        data, address = sniffer.recvfrom(65535)
        print("Source IP:", address[0])    print("Packet
Size:", len(data), "bytes")    print("-" * 40)
except KeyboardInterrupt:
    print("\nPacket Sniffer Stopped")

finally:
    sniffer.close()
```

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A screenshot of a Windows desktop environment. The primary focus is the Visual Studio Code (VS Code) application. The Explorer sidebar on the left shows a project named 'CN-14' with a file 'packet_sniffer.py'. The main editor area displays the code for 'packet_sniffer.py', which is a Python script for a network packet sniffer. The script imports the 'socket' module, sets up a raw socket on a specific host and port, and enters a loop to receive and print incoming packet data. The status bar at the bottom of the editor indicates the file is encoded in UTF-8. Below the editor is a terminal window with the command prompt 'PS C:\Users\Santhosha> python .\packet_sniffer.py'. The terminal output shows the script running successfully, listening on IP 10.51.53.83, and capturing three packets from source IP 20.207.73.85, each 40 bytes in size. To the right of the terminal is a 'Chat' sidebar for Copilot, which includes a tip to customize loading messages and a list of recent chat sessions, including 'packet_sniffer.py'. The Windows taskbar at the bottom shows the Start button, a search bar, and several pinned application icons. The system tray in the bottom right corner displays the date and time as '22:37 06-10-2023'.

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```
File Edit Selection View Go Run Terminal Help
CN-14 [Administrator]

Explorer
CN-14
packet_sniffer.py

Problems Output Terminal Ports Debug Console

PS C:\Users\SanthoshA\OneDrive\Documents\CN-14> python .\packet_sniffer.py
Packet sniffer started
Listening on: 10.51.53.83
Press Ctrl+C to stop
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 20.207.73.85
Packet Size: 40 bytes
-----
Source IP: 10.51.53.38
Packet Size: 289 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 104.18.39.21
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 150.171.110.86
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Source IP: 140.82.113.21
Packet Size: 40 bytes
-----
Packet Sniffer Stopped
```